

Improving Preconditioners for LQCD

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Preconditioners for Wilson Fermions

- One of the most computationally expensive parts of a lattice QCD calculation is generating *propagators*, created by solving a linear system with the lattice Dirac operator D , from the discretization of the quark and gluon fields
- As the mass of the light quarks approaches their physical value, the spectrum of the Dirac operator gets closer to the origin, drastically slowing down the converge of iterative solvers
- Different preconditioning techniques are used to accelerate the iterative solution with the Dirac operator:
 - Deflation methods: exact deflation, approximate deflation (multigrid)
 - Even-odd preconditioning
 - Approximate inverses: block Jacobi, domain decomposition, Incomplete LU factorization

New Multigrid upgrade: Chiral SVD extension for prolongators

New *Chiral SVD* prolongators [1]:

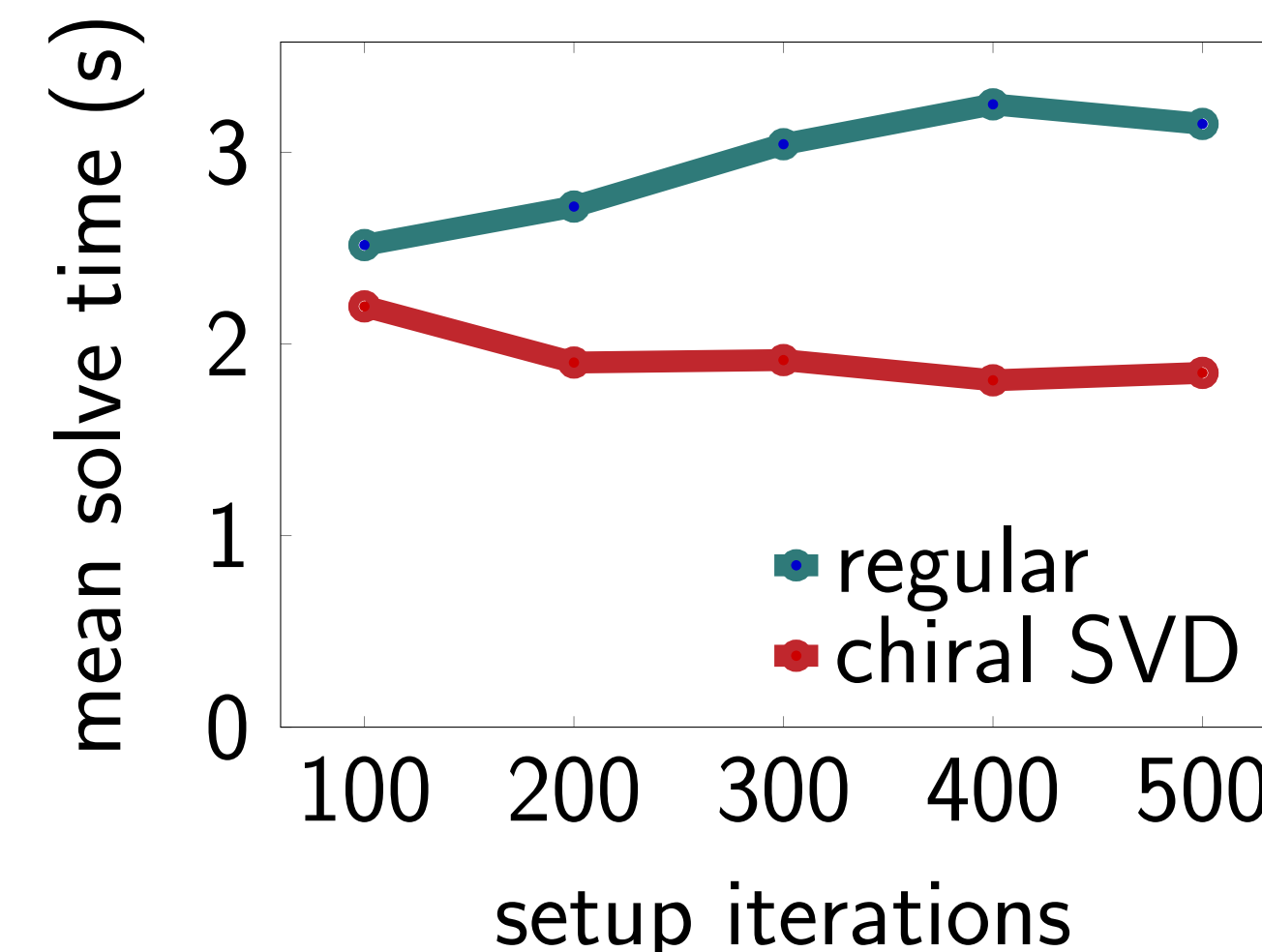
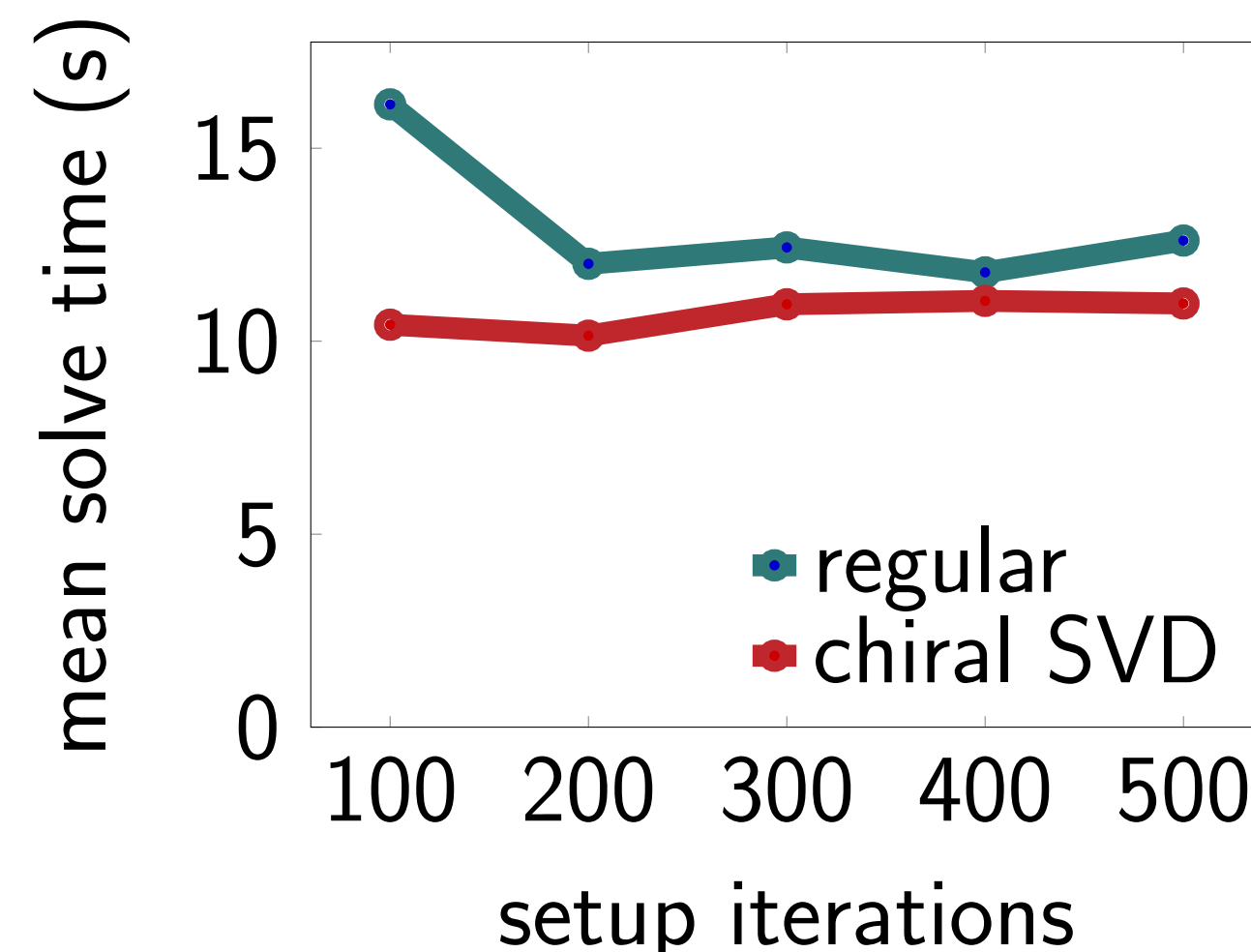
- 1 Get a rough rank K approximation, Ψ , of left singular vectors of D with the smallest singular values, e.g., solving $D\phi = 0$ with K random initial guesses, aka null vectors
- 2 Take each chiral component, $\Psi_0 = (1 + \gamma_5)\Psi$ and $\Psi_1 = (1 - \gamma_5)\Psi$
- 3 Divide the lattice in d small domains, Λ_j , and restrict Ψ_i to each one and select the $k \leq K$ most frequent directions, $\text{SVD}_k(\Psi_i(\Lambda_j))$, e.g., selecting the k left singular vectors with the largest singular values
- 4 Construct the prolongator as $[\text{SVD}_k(\Psi_i(\Lambda_j)) \text{ for } i \in \{0,1\}, 1 \leq j \leq d]$

- The Chiral SVD prolongators are equivalent to the regular prolongators when $k = K$
- For $k < K$, the resulting multigrid preconditioner is better than with the regular prolongators generated with k null vectors

Average number of iterations for each multigrid level on a $32^3 \times 64$ lattice with $K = 2k$ for all levels:

MG Level	Regular Prolong.	CSVD Prolong.
0	12.0	8.0
1	34.0	16.0
2	115.2	58.1

Mean solve time comparing the regular multigrid setup and the Chiral SVD setup, starting by solving $D\phi = 0$ iteratively with different maximum number of iterations at all levels. Tested on an anisotropic $32^3 \times 256$ lattices with $m_\pi \approx 239$ MeV (left) and an isotropic $32^3 \times 64$ lattices with $m_\pi \approx 220$ MeV (right).



[1] T. Whyte et al., *Accelerating multigrid with streaming chiral SVD for Wilson Fermions in lattice QCD*. arXiv

<https://arxiv.org/abs/2505.14399>

Accelerating Multigrid with Approximate Inverse Preconditioners

- Multigrid is usually combined with another preconditioner, especially to accelerate the smoothing and the solution of the coarsest operator
- $$K_{\text{MG}}[A]x = Vy + \underbrace{\text{solve}(A, x - AVy)}_{\text{smoothing}}, \quad \text{where } y = \text{solve}(\underbrace{V^\dagger AV}_{\text{coarse op.}}, V^\dagger x)$$
- We are exploring the potential of using Incomplete LU (ILU) factorizations as preconditioners as complement for multigrid
 - Despite the fact that solving the triangular factors is difficult to parallelize, ILU may be useful:
 - as part of the domain decomposition based preconditioners
 - for solving the coarse operator with a small number of processes
 - We are planning on taking advantage of SuperLU optimized sparse triangular solver kernels

Preliminary study of several preconditioners for accelerating the solution of the coarsest operator on a typical 2-level multigrid for a lattice of $32^4 \times 64$.

